Week 13 Report

Computational Statistics for Data Science · ISS507700

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I have completed most of the work by myself, but I used AI tools to help with debugging, understanding some concepts, and checking for typos.

Q0 · Setup

```
library(readxl)
library(ggplot2)
library(psych)

file_name <- "security_questions.xlsx"

security <- read_excel(file_name, sheet = "data")
security <- as.data.frame(security)

# make sure all Q columns are numeric
security[] <- lapply(security, as.numeric)

# remove rows with missing values, if any
security <- na.omit(security)

# question wording sheet
questions <- read_excel(file_name, sheet = "questions", col_names = FALSE)
questions <- as.data.frame(questions)

colnames(questions) <- c("Item", "Question")
```

Q1 · Perform a parallel analysis

(a) Show a single visualization with:

(i) scree plot of original data's eigen decomposition

(ii) scree plot of simulated noise (from 1000 noise samples) as:

1. ribbon of 90% CI of noise eigenvalues
2. two lines for mean and 95th percentile of noise eigenvalues

3. a horizontal line showing the eigenvalue = 1 cutoff.

```
library(readxl)
library(ggplot2)

# load data
security <- read_excel("security_questions.xlsx", sheet = "data")
security <- as.data.frame(security)

# make sure all items are numeric
security[] <- lapply(security, as.numeric)

# remove rows with missing values
security <- na.omit(security)

# original data eigenvalues
original_eigen <- eigen(cor(security))$values

# function to simulate random noise eigenvalues
sim_noise_ev <- function(n, p) {
  noise <- data.frame(replicate(p, rnorm(n)))
  eigen(cor(noise))$values
}

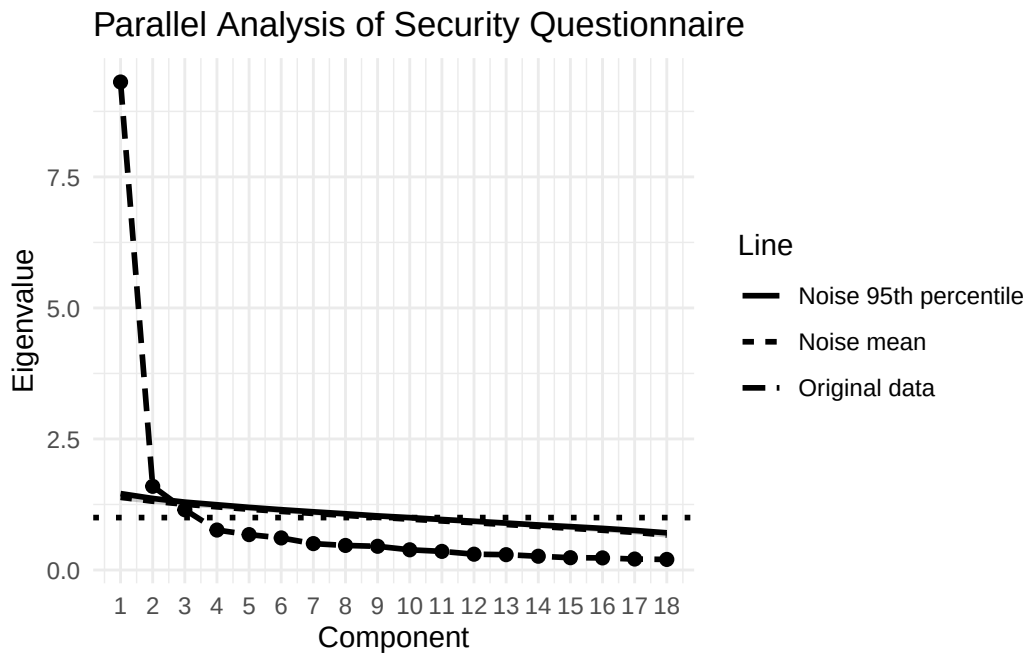
set.seed(42)

# 1000 simulated noise samples
noise_eigen <- replicate(
  1000,
  sim_noise_ev(n = nrow(security), p = ncol(security))
)

# summarize noise eigenvalues
parallel_df <- data.frame(
  component = 1:ncol(security),
  original = original_eigen,
  noise_mean = apply(noise_eigen, 1, mean),
  noise_05 = apply(noise_eigen, 1, quantile, probs = 0.05),
  noise_95 = apply(noise_eigen, 1, quantile, probs = 0.95)
)

# visualization
ggplot(parallel_df, aes(x = component)) +
  geom_ribbon(aes(ymin = noise_05, ymax = noise_95), alpha = 0.25) +
  geom_line(aes(y = original, linetype = "Original data"), linewidth = 1) +
  geom_point(aes(y = original), size = 2) +
  geom_line(aes(y = noise_mean, linetype = "Noise mean"), linewidth = 1) +
  geom_line(aes(y = noise_95, linetype = "Noise 95th percentile"), linewidth = 1) +
```

```
  geom_hline(yintercept = 1, linetype = "dotted", linewidth = 1) +
  scale_x_continuous(breaks = 1:ncol(security)) +
  labs(
    title = "Parallel Analysis of Security Questionnaire",
    x = "Component",
    y = "Eigenvalue",
    linetype = "Line"
  ) +
  theme_minimal()
```



The visualization compares the original eigenvalues with simulated-noise eigenvalues from 1000 random datasets.

The original data line drops sharply after the first component, while the simulated-noise mean and 95th percentile lines provide reference values for what eigenvalues could look like by chance. The horizontal eigenvalue = 1 line also shows the traditional Kaiser cutoff.

(b) How many dimensions would you retain if we used Parallel Analysis?

```
# retain components whose original eigenvalues are larger than noise mean
retain_mean <- sum(parallel_df$original > parallel_df$noise_mean)

# stricter rule: retain components whose original eigenvalues are larger than noise 95th percent
retain_95 <- sum(parallel_df$original > parallel_df$noise_95)

retain_mean
```

```
#> [1] 2
```

```
retain_95
```

```
#> [1] 2
```

Using Parallel Analysis, I would retain 2 dimensions.

Both the noise-mean rule and the stricter 95th-percentile rule give 2 retained components in my output. This means only the first two original eigenvalues are clearly larger than what we would expect from random noise.

Q2 · Treat the underlying dimensions of the security dataset as factors and examine factor loadings, using the principal() method from the psych package

---

(a) Looking at the loadings of the first 3 principal components, to which components does each item seem to best belong?

```
# extract 3 unrotated principal components
pc3 <- principal(
  r = security,
  nfactors = 3,
  rotate = "none",
  scores = TRUE
)

# print loading table
print(pc3, cut = 0.30, sort = FALSE)
```

```
#> Principal Components Analysis
#> Call: principal(r = security, nfactors = 3, rotate = "none", scores = TRUE)
#> Standardized loadings (pattern matrix) based upon correlation matrix
#>      PC1   PC2   PC3   h2   u2 com
#> Q1  0.82             0.69 0.31 1.1
#> Q2  0.67             0.46 0.54 1.0
#> Q3  0.77             0.60 0.40 1.0
#> Q4  0.62  0.64       0.81 0.19 2.1
#> Q5  0.69       -0.54 0.77 0.23 1.9
#> Q6  0.68             0.52 0.48 1.2
#> Q7  0.66 -0.32  0.32 0.64 0.36 2.0
#> Q8  0.79       -0.34 0.74 0.26 1.4
#> Q9  0.72             0.62 0.38 1.4
#> Q10 0.69       -0.53 0.76 0.24 1.9
#> Q11 0.75             0.66 0.34 1.4
#> Q12 0.63  0.64       0.82 0.18 2.1
#> Q13 0.71             0.52 0.48 1.0
#> Q14 0.81             0.69 0.31 1.1
```

```
#> Q15 0.70       -0.33 0.61 0.39 1.4
#> Q16 0.76             0.65 0.35 1.3
#> Q17 0.62  0.66       0.83 0.17 2.0
#> Q18 0.81             0.67 0.33 1.1
#>
#>                        PC1  PC2  PC3
#> SS loadings           9.31 1.60 1.15
#> Proportion Var        0.52 0.09 0.06
#> Cumulative Var        0.52 0.61 0.67
#> Proportion Explained  0.77 0.13 0.10
#> Cumulative Proportion 0.77 0.90 1.00
#>
#> Mean item complexity =  1.5
#> Test of the hypothesis that 3 components are sufficient.
#>
#> The root mean square of the residuals (RMSR) is  0.03
#>  with the empirical chi square  129.33  with prob <  0.035
#>
#> Fit based upon off diagonal values = 1
```

```r
# convert loadings to data frame
pc_loadings <- as.data.frame(unclass(pc3$loadings[, 1:3]))

# find component with largest absolute loading for each item
abs_pc_loadings <- abs(pc_loadings)

item_best_pc <- data.frame(
  Item = rownames(pc_loadings),
  pc_loadings,
  Best_Component = colnames(abs_pc_loadings)[max.col(abs_pc_loadings)],
  Best_Loading = apply(pc_loadings, 1, function(x) x[which.max(abs(x))])
)

item_best_pc
```

```
#>     Item       PC1         PC2          PC3 Best_Component Best_Loading
#> Q1    Q1 0.8169846 -0.13941235 -0.002115927            PC1    0.8169846
#> Q2    Q2 0.6726084 -0.01375526  0.089174403            PC1    0.6726084
#> Q3    Q3 0.7655215 -0.03269651  0.089686106            PC1    0.7655215
#> Q4    Q4 0.6233733  0.64307826  0.108031860            PC2    0.6430783
#> Q5    Q5 0.6900841 -0.03126466 -0.542354570            PC1    0.6900841
#> Q6    Q6 0.6828029 -0.10462094  0.207232000            PC1    0.6828029
#> Q7    Q7 0.6566249 -0.31763196  0.324176779            PC1    0.6566249
#> Q8    Q8 0.7861054  0.04235983 -0.343212951            PC1    0.7861054
#> Q9    Q9 0.7230295 -0.23164618  0.203556038            PC1    0.7230295
#> Q10  Q10 0.6861529 -0.09868038 -0.532678749            PC1    0.6861529
#> Q11  Q11 0.7529735 -0.26100673  0.172516196            PC1    0.7529735
#> Q12  Q12 0.6303505  0.63753124  0.121522834            PC2    0.6375312
```

```
#> Q13  Q13 0.7119085 -0.06463837  0.084335919           PC1    0.7119085
#> Q14  Q14 0.8114677 -0.09970016  0.156787046           PC1    0.8114677
#> Q15  Q15 0.7040428  0.01057936 -0.332546876           PC1    0.7040428
#> Q16  Q16 0.7575616 -0.20281591  0.183170175           PC1    0.7575616
#> Q17  Q17 0.6175336  0.66426051  0.110061160           PC2    0.6642605
#> Q18  Q18 0.8067284 -0.11360432 -0.065189145           PC1    0.8067284
```

Most items best belong to PC1, while Q4, Q12, and Q17 best belong to PC2.

I assigned each item to the component where it had the largest absolute loading. Based on this rule, most items best belong to PC1, while Q4, Q12, and Q17 best belong to PC2. No item best belongs to PC3. This suggests that the unrotated first PC acts like a broad general security component, so the unrotated solution is not very easy to interpret item-by-item.

(b) How much of the total variance of the security dataset do the first 3 PCs capture?

```
# variance explained by each PC
pc3$Vaccounted
```

```
#>                            PC1        PC2        PC3
#> SS loadings           9.3109533 1.59633195 1.14955822
#> Proportion Var        0.5172752 0.08868511 0.06386435
#> Cumulative Var        0.5172752 0.60596029 0.66982464
#> Proportion Explained  0.7722546 0.13240049 0.09534487
#> Cumulative Proportion 0.7722546 0.90465513 1.00000000
```

```
# cleaner table
variance_pc3 <- data.frame(
  Component = colnames(pc_loadings),
  SS_Loadings = pc3$Vaccounted["SS loadings", 1:3],
  Proportion_Variance = pc3$Vaccounted["Proportion Var", 1:3],
  Cumulative_Variance = pc3$Vaccounted["Cumulative Var", 1:3]
)

variance_pc3
```

```
#>     Component SS_Loadings Proportion_Variance Cumulative_Variance
#> PC1       PC1    9.310953           0.51727518            0.5172752
#> PC2       PC2    1.596332           0.08868511            0.6059603
#> PC3       PC3    1.149558           0.06386435            0.6698246
```

```
# direct answer for total variance captured by first 3 PCs
total_variance_pc3 <- pc3$Vaccounted["Cumulative Var", 3]
total_variance_pc3
```

```
#> [1] 0.6698246
```

The first 3 PCs capture 66.98 percent of the total variance.

PC1 captures about 51.73%, PC2 captures about 8.87%, and PC3 captures about 6.39%. Together, their cumulative variance is 0.6698, meaning the first three components explain about two-thirds of the security dataset's total variance.

(c) Looking at commonality and uniqueness, which items are less than adequately explained by the first 3 principal components?

```
# add commonality and uniqueness to item table
item_explanation <- data.frame(
  Item = rownames(pc_loadings),
  h2 = pc3$communality,
  u2 = pc3$uniquenesses
)

item_explanation
```

```
#>     Item        h2        u2
#> Q1    Q1 0.6869041 0.3130959
#> Q2    Q2 0.4605433 0.5394567
#> Q3    Q3 0.5951359 0.4048641
#> Q4    Q4 0.8138147 0.1861853
#> Q5    Q5 0.7713420 0.2286580
#> Q6    Q6 0.5201104 0.4798896
#> Q7    Q7 0.6371369 0.3628631
#> Q8    Q8 0.7375512 0.2624488
#> Q9    Q9 0.6178667 0.3821333
#> Q10  Q10 0.7642903 0.2357097
#> Q11  Q11 0.6648554 0.3351446
#> Q12  Q12 0.8185557 0.1814443
#> Q13  Q13 0.5181043 0.4818957
#> Q14  Q14 0.6930021 0.3069979
#> Q15  Q15 0.6063756 0.3936244
#> Q16  Q16 0.6485852 0.3514148
#> Q17  Q17 0.8347032 0.1652968
#> Q18  Q18 0.6679663 0.3320337
```

```
# items with low commonality
less_explained_items <- subset(item_explanation, h2 < 0.50)

less_explained_items
```

```
#>    Item        h2        u2
#> Q2   Q2 0.4605433 0.5394567
```

Using h2 < 0.50 as the cutoff, Q2 is the only item less than adequately explained by the first 3 PCs.

Q2 has commonality h2 $= 0.4605$ and uniqueness u2 $= 0.5395$, meaning less than half of its variance is explained by the three-component solution. All other items have commonality above 0.50.

(d) How many measurement items share similar loadings between 2 or more components?

```
# Full unrotated loading matrix
pc_loadings <- as.data.frame(unclass(pc3$loadings[, 1:3]))

# Absolute loadings because we care about loading strength
abs_pc_loadings <- abs(pc_loadings)

# Find strongest and second strongest loading for each item
loading_similarity <- data.frame(
  Item = rownames(pc_loadings),
  PC1 = pc_loadings$PC1,
  PC2 = pc_loadings$PC2,
  PC3 = pc_loadings$PC3,
  Strongest_Abs_Loading = apply(abs_pc_loadings, 1, max),
  Second_Abs_Loading = apply(abs_pc_loadings, 1, function(x) sort(x, decreasing = TRUE)[2])
)

# Difference between strongest and second strongest loading
loading_similarity$Difference <-
  loading_similarity$Strongest_Abs_Loading - loading_similarity$Second_Abs_Loading

similar_loading_items <- subset(
  loading_similarity,
  Second_Abs_Loading >= 0.30 & Difference <= 0.20
)

similar_loading_items
```

```
#>     Item       PC1         PC2        PC3 Strongest_Abs_Loading
#> Q4    Q4 0.6233733  0.64307826  0.1080319             0.6430783
#> Q5    Q5 0.6900841 -0.03126466 -0.5423546             0.6900841
#> Q10  Q10 0.6861529 -0.09868038 -0.5326787             0.6861529
#> Q12  Q12 0.6303505  0.63753124  0.1215228             0.6375312
#> Q17  Q17 0.6175336  0.66426051  0.1100612             0.6642605
#>     Second_Abs_Loading  Difference
#> Q4           0.6233733 0.019705008
#> Q5           0.5423546 0.147729540
#> Q10          0.5326787 0.153474160
#> Q12          0.6303505 0.007180715
#> Q17          0.6175336 0.046726904
```

```
nrow(similar_loading_items)
```

```
#> [1] 5
```

```
strict_similar_loading_items <- subset(
  loading_similarity,
  Second_Abs_Loading >= 0.70
```

```
)

strict_similar_loading_items
```

```
#> [1] Item                  PC1                   PC2
#> [4] PC3                   Strongest_Abs_Loading Second_Abs_Loading
#> [7] Difference
#> <0 rows> (or 0-length row.names)
```

```
nrow(strict_similar_loading_items)
```

```
#> [1] 0
```

There are 5 items that share similar loadings between 2 or more components: Q4, Q5, Q10, Q12, and Q17.

Using my exploratory rule, 5 items have similar loadings across two components: Q4, Q5, Q10, Q12, and Q17. However, according to the slide standard that $\lambda > 0.70$ is a strong loading, none of these items have strong loadings on two or more components. Therefore, the answer depends on whether we mean "similar moderate loadings" or "similar strong loadings."

(e) Can you interpret a 'meaning' behind the first principal component from the items that load best upon it? (see the wording of the questions of those items)

```
# items sorted by strength of PC1 loading
pc1_interpretation <- data.frame(
  Item = rownames(pc_loadings),
  PC1_Loading = pc_loadings$PC1,
  Abs_PC1_Loading = abs(pc_loadings$PC1)
)

pc1_interpretation <- merge(
  pc1_interpretation,
  questions,
  by = "Item",
  all.x = TRUE
)

pc1_interpretation <- pc1_interpretation[
  order(pc1_interpretation$Abs_PC1_Loading, decreasing = TRUE),
]

library(knitr)

pc1_interpretation$Question_wrapped <- sapply(
  pc1_interpretation$Question,
  function(x) paste(strwrap(x, width = 50), collapse = "<br>")
)
```

```
kable(
  pc1_interpretation[, c("Item", "PC1_Loading", "Question")],
  col.names = c("Item", "PC1 Loading", "Question")
)
```

	Item	PC1 Loading	Question
1	Q1	0.8169846	I am convinced that this site respects the confidentiality of the transactions received from me
6	Q14	0.8114677	This site devotes time and effort to verify the accuracy of the information in transit
10	Q18	0.8067284	This site uses some security controls for the confidentiality of the transactions received from me
17	Q8	0.7861054	This site ascertains my identity before processing the transactions received from me
12	Q3	0.7655215	This site checks the information communicated with me for accuracy
8	Q16	0.7575616	Databases that contain my personal information are protected from unauthorized access
3	Q11	0.7529735	This site devotes time and effort to preventing unauthorized access to my personal information
18	Q9	0.7230295	I can remove my personal information from this site when I want to
5	Q13	0.7119085	This site provides me with some evidence to protect against its denial of having sent a message
7	Q15	0.7040428	This site ascertains my identity before sending any messages to me
14	Q5	0.6900841	The transactions I send are transmitted to the real site to which I want to transmit
2	Q10	0.6861529	The messages I receive are transmitted from the real site from which I want to receive them
15	Q6	0.6828029	This site checks all communications between the site and me for protection from wiretapping or eavesdropping
11	Q2	0.6726084	All communications with this site are restricted to the site and me
16	Q7	0.6566249	This site never sells my personal information in their computer databases to other companies
4	Q12	0.6303505	This site takes steps to make sure that the information in transit is not deleted
13	Q4	0.6233733	This site provides me with some evidence to protect against its denial of having received a transaction from me
9	Q17	0.6175336	This site provides me with some evidence to protect against its denial of having participated in a transaction after processing it

The first principal component seems to represent a general sense of website security, privacy, and trustworthiness.

The strongest PC1 items involve confidentiality, accuracy, identity verification, protection of personal information, restricted communication, and secure transactions. Since many different security-related items load strongly on PC1, it appears to capture an overall perceived security factor rather than one narrow security feature.

Q3 · To improve interpretability of loadings, rotate principal component axes using the varimax technique to get rotated components (RCs; extract and rotate only three principal components)

(a) Does each RC explain the same or different amount of individual variance than the original PCs?

```
# original 3 unrotated PCs
pc3 <- principal(
  r = security,
  nfactors = 3,
  rotate = "none",
  scores = TRUE
)

# 3 varimax rotated components
rc3 <- principal(
  r = security,
  nfactors = 3,
  rotate = "varimax",
  scores = TRUE
)

# compare variance explained
variance_compare_3 <- data.frame(
  Component = c("1", "2", "3"),
  PC_SS_Loadings = pc3$Vaccounted["SS loadings", 1:3],
  PC_Proportion = pc3$Vaccounted["Proportion Var", 1:3],
  RC_SS_Loadings = rc3$Vaccounted["SS loadings", 1:3],
  RC_Proportion = rc3$Vaccounted["Proportion Var", 1:3]
)

variance_compare_3
```

```
#>     Component PC_SS_Loadings PC_Proportion RC_SS_Loadings RC_Proportion
#> PC1         1       9.310953    0.51727518       5.613148     0.3118416
#> PC2         2       1.596332    0.08868511       3.490139     0.1938966
#> PC3         3       1.149558    0.06386435       2.953556     0.1640864
```

Each rotated component explains a different amount of individual variance than the original PCs.

Before rotation, PC1 explains much more variance than PC2 and PC3. After varimax rotation, the variance is redistributed more evenly across RC1, RC2, and RC3, with proportions of about 31.18%, 19.39%, and 16.41%.

(b) Do the three RCs explain the same or different amount of cumulative variance as the three PCs ?

```
pc3_total_variance <- pc3$Vaccounted["Cumulative Var", 3]
rc3_total_variance <- rc3$Vaccounted["Cumulative Var", 3]

pc3_total_variance
```

```
#> [1] 0.6698246
```

```
rc3_total_variance
```

```
#> [1] 0.6698246
```

```
# check if they are practically the same
round(pc3_total_variance, 4) == round(rc3_total_variance, 4)
```

```
#> [1] TRUE
```

> The three RCs explain the same cumulative variance as the three original PCs: 66.98 percent.

Rotation changes how variance is distributed across the components, but it does not change the total variance explained by the full set of extracted components. Both the unrotated and rotated three-component solutions have cumulative variance 0.6698.

(c) Looking back at the items that shared similar loadings with multiple principal components (#2d), do those items have more clearly differentiated loadings among rotated components?

```
# unrotated loadings
pc_loadings <- as.data.frame(unclass(pc3$loadings[, 1:3]))
abs_pc_loadings <- abs(pc_loadings)

# find items with similar loadings in unrotated PCs
loading_similarity_pc <- data.frame(
  Item = rownames(pc_loadings),
  pc_loadings,
  Strongest_PC = apply(abs_pc_loadings, 1, max),
  Second_PC = apply(abs_pc_loadings, 1, function(x) sort(x, decreasing = TRUE)[2])
)

loading_similarity_pc$Difference_PC <-
  loading_similarity_pc$Strongest_PC - loading_similarity_pc$Second_PC

similar_loading_items_pc <- subset(
  loading_similarity_pc,
  Second_PC >= 0.30 & Difference_PC <= 0.20
)
```

```
similar_loading_items_pc
```

```
#>     Item       PC1         PC2        PC3 Strongest_PC Second_PC Difference_PC
#> Q4    Q4 0.6233733  0.64307826  0.1080319    0.6430783 0.6233733   0.019705008
#> Q5    Q5 0.6900841 -0.03126466 -0.5423546    0.6900841 0.5423546   0.147729540
#> Q10  Q10 0.6861529 -0.09868038 -0.5326787    0.6861529 0.5326787   0.153474160
#> Q12  Q12 0.6303505  0.63753124  0.1215228    0.6375312 0.6303505   0.007180715
#> Q17  Q17 0.6175336  0.66426051  0.1100612    0.6642605 0.6175336   0.046726904
```

```
rc_loadings_only <- as.data.frame.matrix(unclass(rc3$loadings[, 1:3]))
rc_loadings_only$Item <- rownames(rc_loadings_only)

rotated_cross_items <- rc_loadings_only[
  rc_loadings_only$Item %in% similar_loading_items_pc$Item,
]

rotated_cross_items
```

```
#>           RC1       RC3       RC2 Item
#> Q4  0.2182880 0.1933627 0.8536838   Q4
#> Q5  0.2441735 0.8279850 0.1617475   Q5
#> Q10 0.2768895 0.8229206 0.1020988  Q10
#> Q12 0.2327616 0.1861745 0.8542346  Q12
#> Q17 0.2054021 0.1869028 0.8703910  Q17
```

Yes, the items that had similar loadings across multiple PCs become more clearly differentiated after varimax rotation.

In the unrotated solution, Q4, Q5, Q10, Q12, and Q17 had close loadings on more than one component. After rotation, these items load more strongly on one rotated component, making their component membership easier to interpret.

(d) Can you now more easily interpret the "meaning" of the 3 rotated components from the items that load best upon each of them? (see the wording of the questions of those items)

```
# Rotated 3-component solution
rc3 <- principal(
  r = security,
  nfactors = 3,
  rotate = "varimax",
  scores = TRUE
)

# Extract rotated loadings safely
rc_loadings <- as.data.frame.matrix(unclass(rc3$loadings[, 1:3]))

# Force item names back into the table
```

```r
rc_loadings$Item <- rownames(rc_loadings)

# Put Item first
rc_loadings <- rc_loadings[, c("Item", "RC1", "RC2", "RC3")]

# Absolute loadings only for comparison
abs_rc_loadings <- abs(rc_loadings[, c("RC1", "RC2", "RC3")])

# Find best rotated component for each item
item_best_rc <- data.frame(
  Item = rc_loadings$Item,
  rc_loadings[, c("RC1", "RC2", "RC3")],
  Best_RC = colnames(abs_rc_loadings)[max.col(abs_rc_loadings)],
  Best_Loading = apply(rc_loadings[, c("RC1", "RC2", "RC3")], 1, function(x) {
    x[which.max(abs(x))]
  })
)

# Add question wording
item_best_rc_questions <- merge(
  item_best_rc,
  questions,
  by = "Item",
  all.x = TRUE
)

# Sort by component and loading strength
item_best_rc_questions <- item_best_rc_questions[
  order(item_best_rc_questions$Best_RC, -abs(item_best_rc_questions$Best_Loading)),
]

library(knitr)

item_best_rc_questions$Question_wrapped <- sapply(
  item_best_rc_questions$Question,
  function(x) paste(strwrap(x, width = 50), collapse = "<br>")
)

kable(
  item_best_rc_questions[, c("Item", "Best_RC", "Best_Loading", "Question")],
  col.names = c("Item", "Best RC", "Best Loading", "Question")
)
```

	Item	Best RC	Best Loading	Question
16	Q7	RC1	0.7895344	This site never sells my personal information in their computer databases to other companies
3	Q11	RC1	0.7573493	This site devotes time and effort to preventing unauthorized access to my personal information
8	Q16	RC1	0.7396241	Databases that contain my personal information are protected from unauthorized access
18	Q9	RC1	0.7378148	I can remove my personal information from this site when I want to
6	Q14	RC1	0.7187578	This site devotes time and effort to verify the accuracy of the information in transit
1	Q1	RC1	0.6602758	I am convinced that this site respects the confidentiality of the transactions received from me
15	Q6	RC1	0.6524225	This site checks all communications between the site and me for protection from wiretapping or eavesdropping
12	Q3	RC1	0.6206018	This site checks the information communicated with me for accuracy
10	Q18	RC1	0.6090325	This site uses some security controls for the confidentiality of the transactions received from me
5	Q13	RC1	0.5931915	This site provides me with some evidence to protect against its denial of having sent a message
11	Q2	RC1	0.5437243	All communications with this site are restricted to the site and me
9	Q17	RC2	0.8703910	This site provides me with some evidence to protect against its denial of having participated in a transaction after processing it
4	Q12	RC2	0.8542346	This site takes steps to make sure that the information in transit is not deleted
13	Q4	RC2	0.8536838	This site provides me with some evidence to protect against its denial of having received a transaction from me
14	Q5	RC3	0.8279850	The transactions I send are transmitted to the real site to which I want to transmit
2	Q10	RC3	0.8229206	The messages I receive are transmitted from the real site from which I want to receive them
17	Q8	RC3	0.7062018	This site ascertains my identity before processing the transactions received from me
7	Q15	RC3	0.6557490	This site ascertains my identity before sending any messages to me

Yes, the 3 rotated components are easier to interpret: RC1 represents general privacy/security protection, RC2 represents transaction integrity/non-repudiation, and RC3 represents authenticity/identity verification.

RC1 contains many items about personal information, confidentiality, secure communication, accuracy, and protection from unauthorized access. RC2 is mainly formed by Q17, Q12, and Q4, which focus on evidence, denial, and making sure information is not deleted. RC3 is mainly formed by Q5, Q10, Q8, and Q15, which focus on communicating with the real site and verifying identity.

(e) If we reduced the number of extracted and rotated components to 2, does the meaning of our rotated components change?

```
# Extract and rotate only 2 components
rc2 <- principal(
  r = security,
  nfactors = 2,
  rotate = "varimax",
  scores = TRUE
)

print(rc2, cut = 0.30, sort = FALSE)
```

```
#> Principal Components Analysis
#> Call: principal(r = security, nfactors = 2, rotate = "varimax", scores = TRUE)
#> Standardized loadings (pattern matrix) based upon correlation matrix
#>      RC1  RC2   h2   u2 com
#> Q1  0.78      0.69 0.31 1.2
#> Q2  0.60 0.31 0.45 0.55 1.5
#> Q3  0.69 0.34 0.59 0.41 1.5
#> Q4       0.86 0.80 0.20 1.1
#> Q5  0.62 0.31 0.48 0.52 1.5
#> Q6  0.65      0.48 0.52 1.3
#> Q7  0.73      0.53 0.47 1.0
#> Q8  0.67 0.42 0.62 0.38 1.7
#> Q9  0.75      0.58 0.42 1.1
#> Q10 0.65      0.48 0.52 1.3
#> Q11 0.79      0.64 0.36 1.1
#> Q12      0.86 0.80 0.20 1.2
#> Q13 0.65      0.51 0.49 1.4
#> Q14 0.76 0.30 0.67 0.33 1.3
#> Q15 0.61 0.35 0.50 0.50 1.6
#> Q16 0.76      0.62 0.38 1.1
#> Q17      0.88 0.82 0.18 1.1
#> Q18 0.76      0.66 0.34 1.3
#>
#>                        RC1  RC2
#> SS loadings           7.52 3.39
#> Proportion Var        0.42 0.19
#> Cumulative Var        0.42 0.61
#> Proportion Explained  0.69 0.31
#> Cumulative Proportion 0.69 1.00
#>
#> Mean item complexity =  1.3
#> Test of the hypothesis that 2 components are sufficient.
#>
#> The root mean square of the residuals (RMSR) is  0.04
#>  with the empirical chi square  219.84  with prob <  3.9e-08
```

```
#>
#> Fit based upon off diagonal values = 0.99
```

```r
# Extract 2-component rotated loadings safely
rc2_loadings <- as.data.frame.matrix(unclass(rc2$loadings[, 1:2]))

# Force item names back
rc2_loadings$Item <- rownames(rc2_loadings)

# Put Item first
rc2_loadings <- rc2_loadings[, c("Item", "RC1", "RC2")]

# Absolute loadings only for comparison
abs_rc2_loadings <- abs(rc2_loadings[, c("RC1", "RC2")])

# Best RC for each item
item_best_rc2 <- data.frame(
  Item = rc2_loadings$Item,
  rc2_loadings[, c("RC1", "RC2")],
  Best_RC = colnames(abs_rc2_loadings)[max.col(abs_rc2_loadings)],
  Best_Loading = apply(rc2_loadings[, c("RC1", "RC2")], 1, function(x) {
    x[which.max(abs(x))]
  })
)

# Add question wording
item_best_rc2_questions <- merge(
  item_best_rc2,
  questions,
  by = "Item",
  all.x = TRUE
)

item_best_rc2_questions <- item_best_rc2_questions[
  order(item_best_rc2_questions$Best_RC, -abs(item_best_rc2_questions$Best_Loading)),
]

item_best_rc2_questions$Question_wrapped <- sapply(
  item_best_rc2_questions$Question,
  function(x) paste(strwrap(x, width = 50), collapse = "<br>")
)

kable(
  item_best_rc2_questions[, c("Item", "Best_RC", "Best_Loading", "Question")],
  col.names = c("Item", "Best RC", "Best Loading", "Question")
)
```

	Item	Best RC	Best Loading	Question
3	Q11	RC1	0.7855784	This site devotes time and effort to preventing unauthorized access to my personal information
1	Q1	RC1	0.7830951	I am convinced that this site respects the confidentiality of the transactions received from me
10	Q18	RC1	0.7616746	This site uses some security controls for the confidentiality of the transactions received from me
8	Q16	RC1	0.7615661	Databases that contain my personal information are protected from unauthorized access
6	Q14	RC1	0.7591295	This site devotes time and effort to verify the accuracy of the information in transit
18	Q9	RC1	0.7451939	I can remove my personal information from this site when I want to
16	Q7	RC1	0.7284256	This site never sells my personal information in their computer databases to other companies
12	Q3	RC1	0.6865878	This site checks the information communicated with me for accuracy
17	Q8	RC1	0.6684679	This site ascertains my identity before processing the transactions received from me
5	Q13	RC1	0.6549937	This site provides me with some evidence to protect against its denial of having sent a message
2	Q10	RC1	0.6488232	The messages I receive are transmitted from the real site from which I want to receive them
15	Q6	RC1	0.6487494	This site checks all communications between the site and me for protection from wiretapping or eavesdropping
14	Q5	RC1	0.6197912	The transactions I send are transmitted to the real site to which I want to transmit
7	Q15	RC1	0.6118654	This site ascertains my identity before sending any messages to me
11	Q2	RC1	0.5960420	All communications with this site are restricted to the site and me
9	Q17	RC2	0.8795921	This site provides me with some evidence to protect against its denial of having participated in a transaction after processing it
13	Q4	RC2	0.8638430	This site provides me with some evidence to protect against its denial of having received a transaction from me
4	Q12	RC2	0.8623433	This site takes steps to make sure that the information in transit is not deleted

Yes, the meaning changes when the solution is reduced from 3 rotated components to 2 rotated components.

With only 2 rotated components, the items that formed RC3 in the 3-component solution mostly merge into RC1, so the authenticity/identity-verification meaning becomes less separate.

(f) (ungraded) Looking back at all our results and analyses of this dataset (from this week and previous), how many components (1-3) do you believe we should extract and analyze to understand the security dataset? Feel free to suggest different answers for different purposes.

```
# Eigenvalues of the correlation matrix
eigen_values <- eigen(cor(security))$values

# Cumulative variance for 1, 2, and 3 components
decision_summary <- data.frame(
  Solution = c("1 component", "2 components", "3 components"),
  Cumulative_Variance = c(
    sum(eigen_values[1]) / length(eigen_values),
    sum(eigen_values[1:2]) / length(eigen_values),
    sum(eigen_values[1:3]) / length(eigen_values)
  )
)

decision_summary
```

```
#>       Solution Cumulative_Variance
#> 1  1 component           0.5172752
#> 2 2 components           0.6059603
#> 3 3 components           0.6698246
```

I would retain 2 components for conservative/statistical dimension reduction, but I would examine 3 components for interpretation.

Parallel Analysis supports 2 components because only the first two original eigenvalues are clearly above the simulated noise reference. However, the 3-component rotated solution explains more total variance, about 66.98%, and gives a cleaner interpretation of the questionnaire structure. Therefore, if the goal is strict dimension reduction, I would choose 2 components; if the goal is understanding the theoretical meanings behind the security items, I would also discuss the 3-component solution.